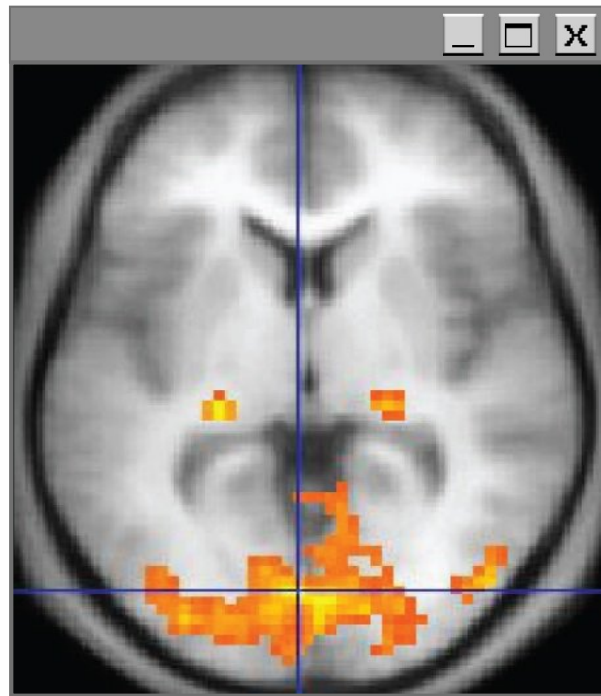


Spatial Resolution in fMRI Neural Decoding

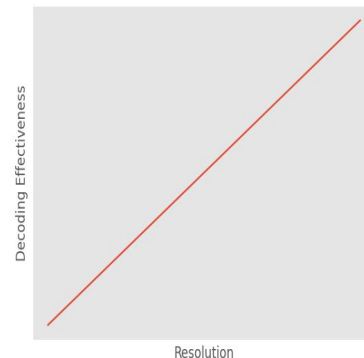
Pod Astarte
Jinghan Huang
Jul 2025



First Question:

What is the relationship between spatial resolution and neural decoding?

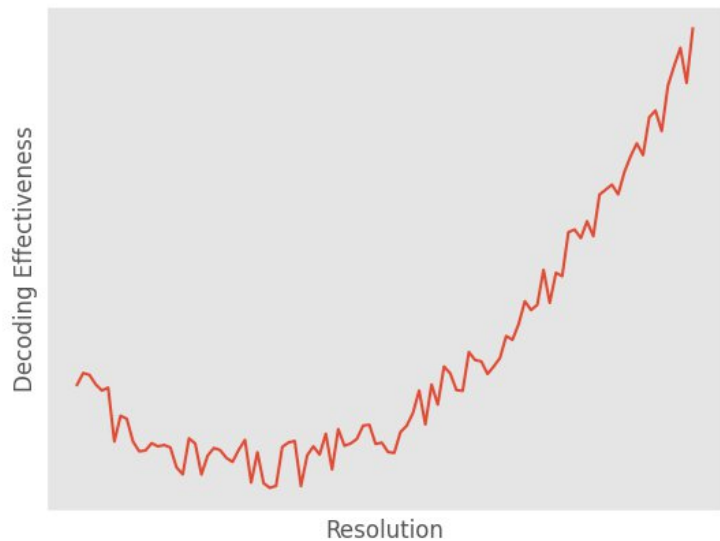
Naive Assumption



The relationship is actually non-linear (Gardumi et al., 2016)

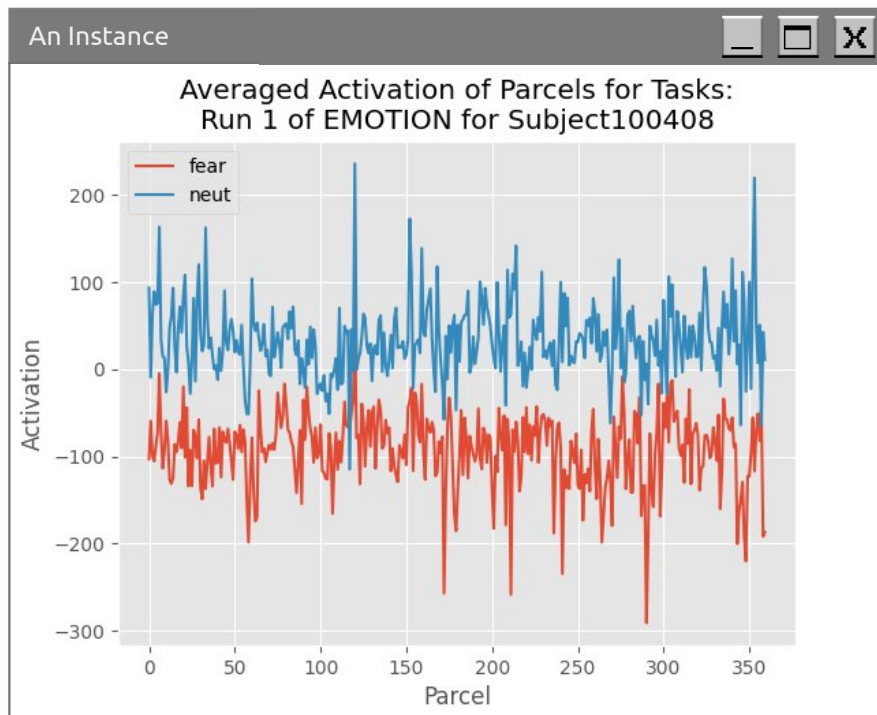
A. Gardumi, D. Ivanov, L. Hausfeld, G. Valente, E. Formisano, and K. Uludağ, "The effect of spatial resolution on decoding accuracy in fMRI multivariate pattern analysis," *NeuroImage*, vol. 132, pp. 32–42, May 2016, doi: 10.1016/j.neuroimage.2016.02.033.

Potential Actual Relationship



Further Questions:

- What is the relationship between spatial resolution and **different types** of neural decoding tasks?
- Can we model the relationship?

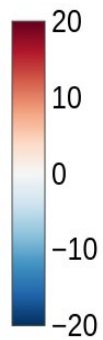
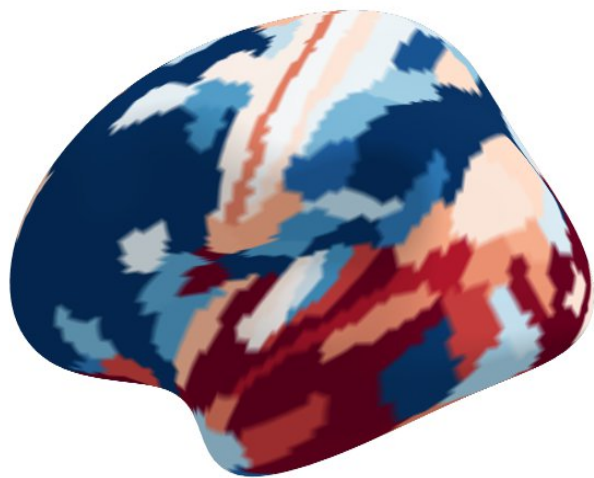


HCP 2021 + Behavior

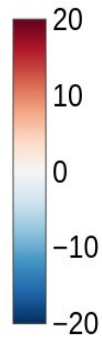
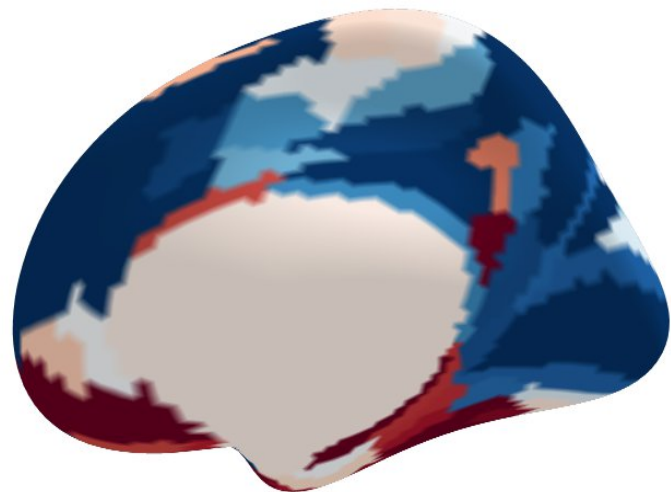
- Subject
 - Experiment [*Motor, WM, Emotion, Gambling, Language, Relational*]
 - Run
 - Data (X) [*3D time-series data*]
 - EVs (Y) [*Multi-class labels*]
 - Metadata

Resolution 1: Parcel

Left Hemisphere

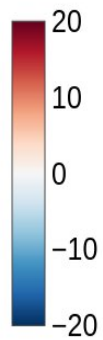
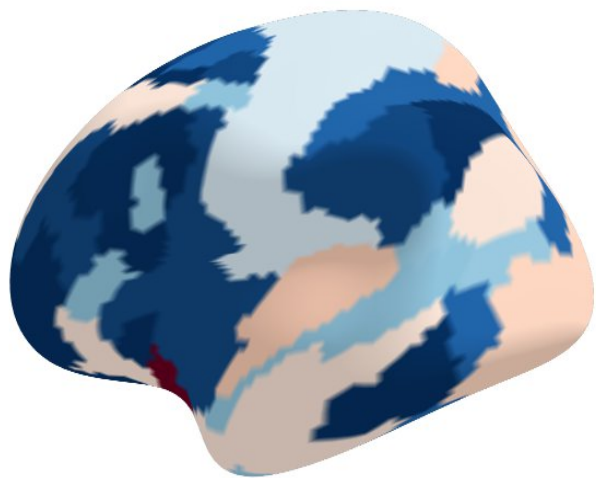


Right Hemisphere

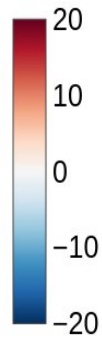
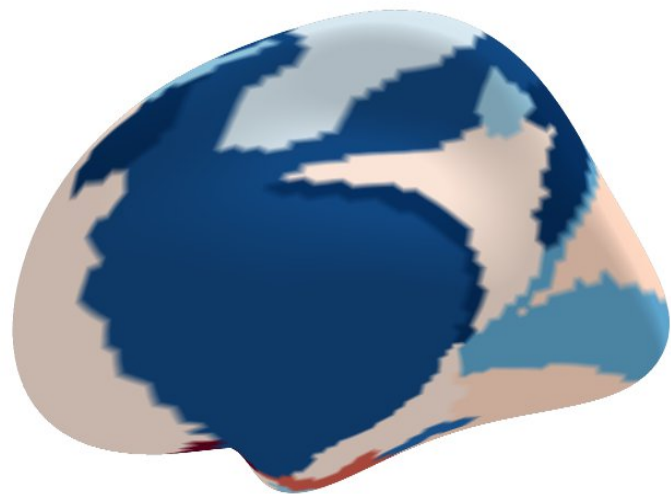


Resolution 2: Network

Left Hemisphere

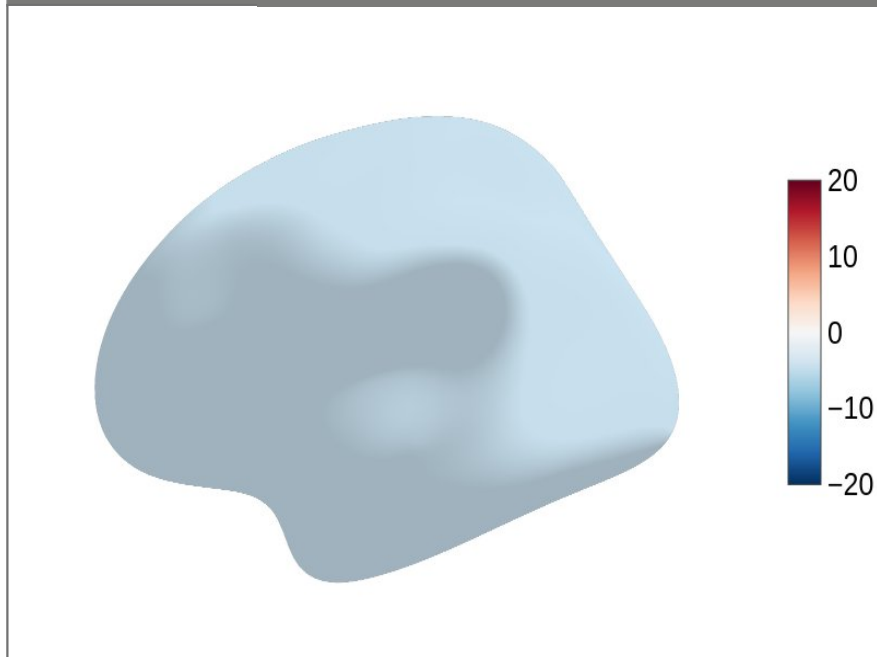


Right Hemisphere

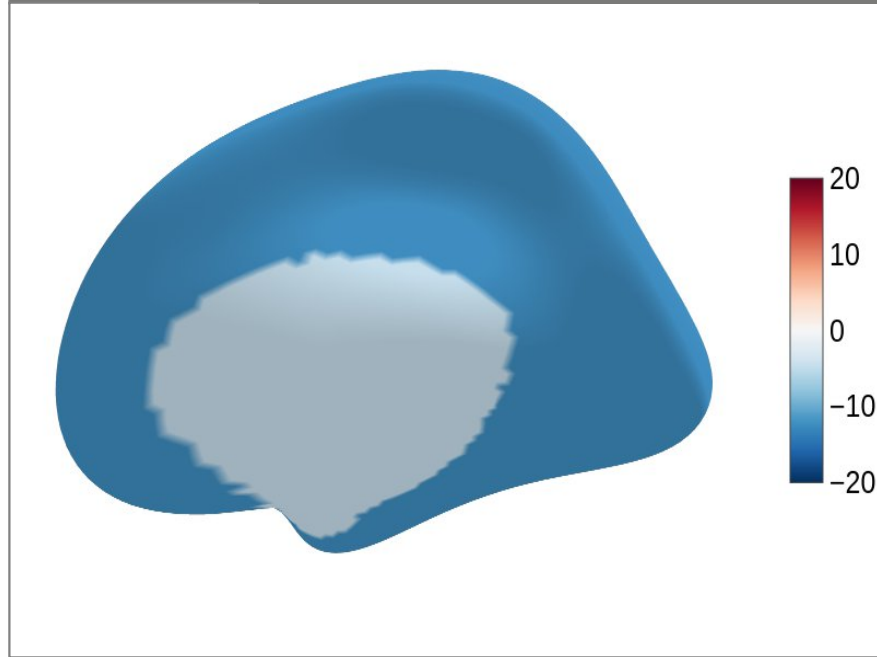


Resolution 3: Hemisphere

Left Hemisphere

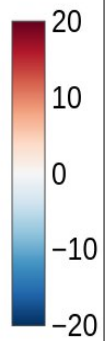
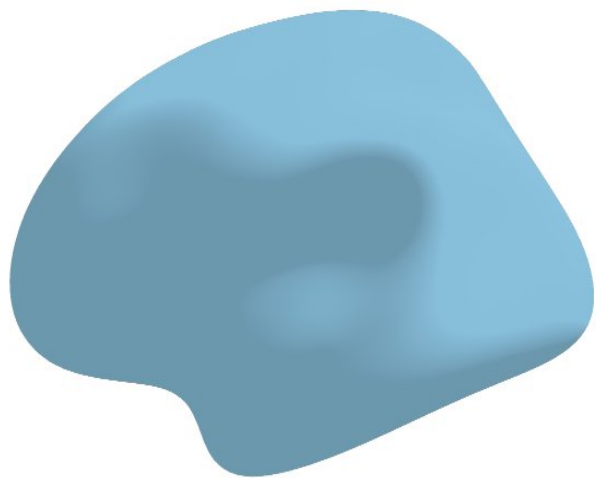


Right Hemisphere

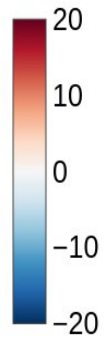
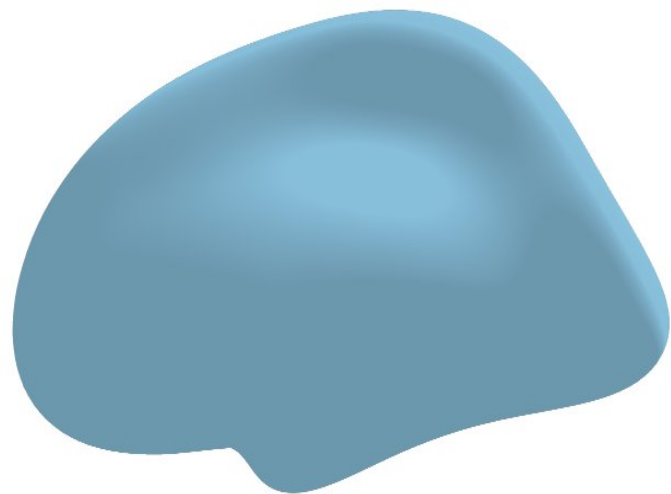


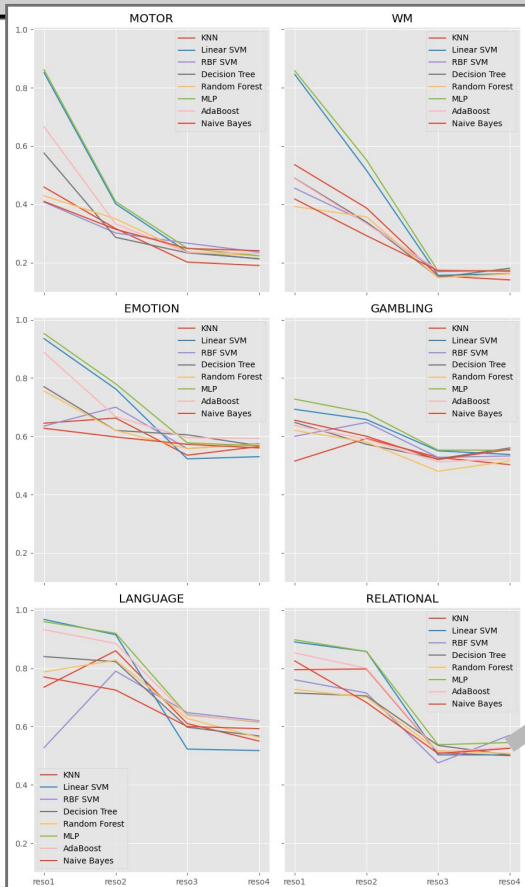
Resolution 4: Cerebral Cortex

Left Hemisphere

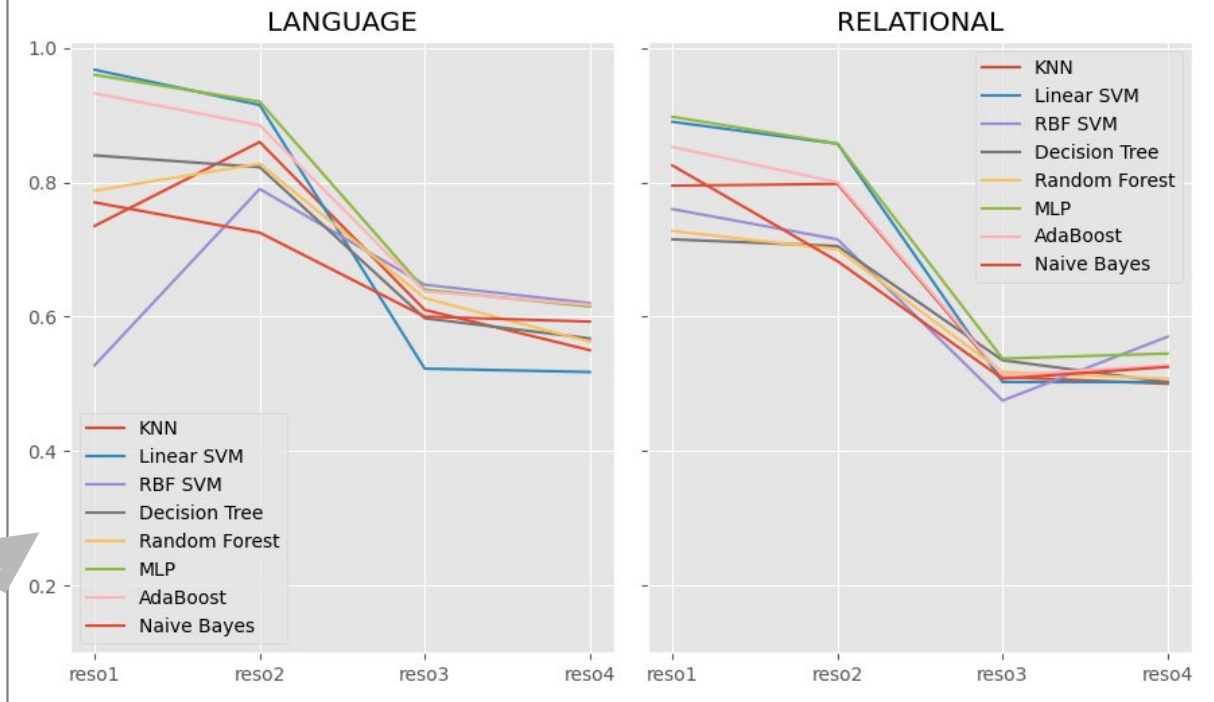


Right Hemisphere





Model-wise Comparison



Outcomes:

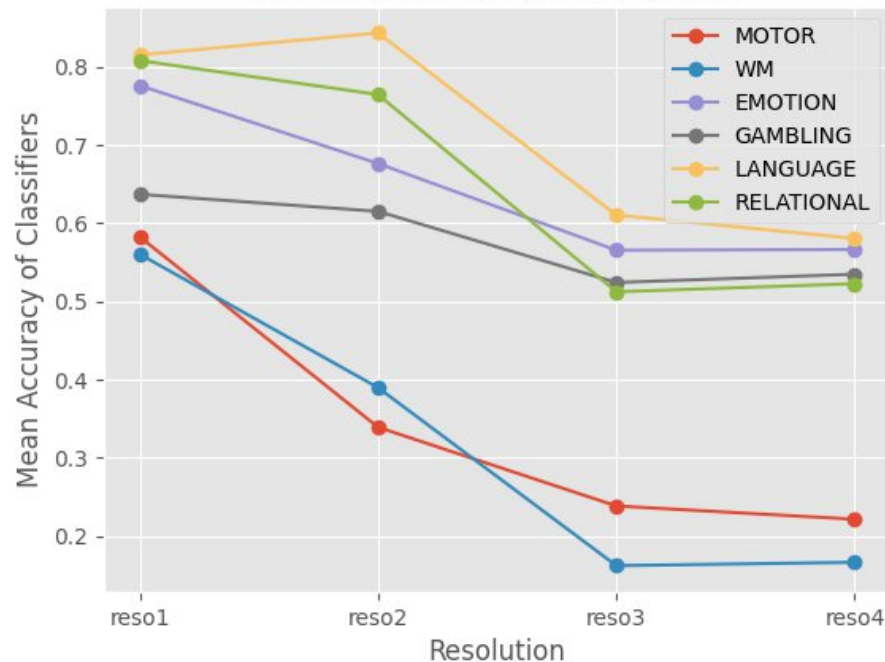
- Non-linear relationship
- Model-wise variability
- Task-wise variability

Future works:

- Quantification of the correlation
- More fine-grained scales

Task-wise Comparison

Spatial Resolution has Varied Non-Linear Impacts on Different Classification Tasks



Thank You

Code available at
<https://github.com/trumpeter122/NMA2025-fMRI-Spatial-Resolution-and-Neural-Decoding.git>